

Engineering 095 - FYRE Pathway 1
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Reflection 2
9/23/2026

What we have done in the past few weeks is really start to work on circuits and the more computer engineering side of this module. At the start we all made little homemade butterfly LED circuits as a little intro to circuits. Then we moved on to more coding basics in python, we made very basic little programs that said things like "Hello, World!" and what our name was and using variables. With these programs we then learned how to utilize Github and make websites for ourselves using AI copilot within github. Our githubs now have all our coding projects with corresponding read me files and a website that has our email and linked in etc, so we can have a working portfolio to start to add to to keep track of all our work and projects for future use. Then we were given arduino circuit boards and worked with breadboards in the Keysight Lab in Packard Lab to make little circuits with different small purposes. We lit a resistor on fire by overloading the voltage and used a rain sensor to detect moisture levels based on voltage. These were all powered by code written by AI with specific prompts directed by us to get us to our complete final result. We also made one with a photoresistor light sensor based on a prototype for a problem given to us where a LED would turn on and off if something was in front of it.

What i have learned from all this circuit and coding experience is how intertwined software and hardware is, there isn't one without the other and some of my gaps in my coding and software knowledge were being filled by hearing and learning about software, it gave me more perspective and opened my mind more to how technology really works fully. I also learned that especially in a tech field like engineering having a portfolio and proof of work and projects done organized and easily accessible is very important for the future in the sense of jobs and internships. You really have to basically brand and promote yourself with a website that's accessible and makes sense and shows who you are and what kind of worker you are, both are important to the job world. Also with all the circuit board stuff I learned a lot more about how those work and how much technology is in everyday things, and honestly how complicated it is. We did pretty trivial labs and examples but there was still so much pre-existing knowledge needed to help you put everything in the right place and bring all the concepts together. However, the hands-on experience we had with them definitely helped me understand and conceptualize what was going on more too.